

Section 4: Expectation and Variance

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1 Introduction

1.1 Logistics

Welcome! The goal of our weekly section is to review content from lecture, with emphasis on big-picture intuition, mathematical derivation, and hands-on practice.

Specifically, we aim for section to be interactive, well-paced, inclusive, and fun! Please do not hesitate to reach out if you have any questions/feedback!

1.2 Office Hours

- See the Google Calendar for the most accurate information!

1.3 Attendance

- bit.ly/stat110harvard

2 Expectation

Idea. Often, we describe a random variable with its *expectation*, which captures the center of its distribution. Even though most random variables can crystallize to a wide range of values, the expectation is a useful statistic to describe the center or long-run average of the distribution.

2.1 Fundamentals

Definition 1 (Expectation). For a random variable X , $\mathbb{E}[X]$ is its *expectation*.

- Informally, $\mathbb{E}[X]$ is the *weighted average* of X , where we weigh each possible *crystallization* by its associated probability.
- Mathematically,

$$\mathbb{E}[X] = \sum_{x \in \text{supp}(X)} \underbrace{x}_{\text{value}} \underbrace{P(X=x)}_{\text{PMF at value}}.$$

- E.g., let $X \sim \text{Bern}(\frac{1}{3})$, with support $\{0, 1\}$. Thus,

$$P(X=0) = \frac{2}{3}, \quad P(X=1) = \frac{1}{3},$$

so

$$\mathbb{E}[X] = \sum_{x \in \{0,1\}} xP(X=x) = \left(0 \times \frac{2}{3}\right) + \left(1 \times \frac{1}{3}\right) = \frac{1}{3}.$$

 **Concept Checker 1.** Let X be the value from rolling a fair 6-sided die. Find $\mathbb{E}[X]$.

Solution

$$\mathbb{E}[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) = 3.5.$$

 **Concept Checker 2.** Why is $\mathbb{E}[X]$ a valid expression but $P(X)$ is not?

Solution

X is a random variable, not an event! Expectation takes a random variable as input and returns a number as output. Probability takes an event as input and returns a number (between 0 and 1) as output.

2.2 Expectation Toolkit

Definition 2 (Linearity of expectation). For random variables X and Y and constants a , b , and c ,

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$$

- We call c a *degenerate* random variable, where it crystallizes to c with probability 1.

Definition 3 (Fundamental bridge). The expectation of an *indicator* for an event A is $P(A)$ —i.e.,

$$\mathbb{E}[\mathbb{1}\{A\}] = P(A).$$

Proof. Let $X \sim \text{Bern}(p)$. Thus, $X = 0$ with probability $1 - p$ and $X = 1$ with probability p . By definition of expectation, $\mathbb{E}[X] = 0(1 - p) + 1(p) = p$, so the expectation of a Bernoulli random variable is p . If $\mathbb{1}\{A\}$ is an indicator for an event A , then $\mathbb{1}\{A\} \sim \text{Bern}(P(A))$. $\square$

Definition 4 (Law of the Unconscious Statistician (LOTUS)). For a random variable X and function $g : \mathbb{R} \rightarrow \mathbb{R}$,

$$\mathbb{E}[g(X)] = \sum_{x \in \text{supp}(X)} g(x)P(X = x).$$

- Notice that we still use the PMF of X to find $\mathbb{E}[g(X)]$!

 **Concept Checker 3.** If we want to find $\mathbb{E}[X^2]$ using LOTUS, what is $g(X)$ and $g(x)$?

Solution

g is the function applied to X within the expectation. Thus, $g(X) = X^2$ —the function applied to the random variable—and $g(x) = x^2$ —the function applied to an arbitrary crystallized value.

3 Variance

Idea. In addition to expectation, *variance* is used to describe a random variable—specifically, its spread/uncertainty.

3.1 Fundamentals

Definition 5 (Variance). For a random variable X , $\text{Var}[X]$ is its *variance*.

- Informally, $\text{Var}[X]$ is a measure of the *spread* of X —i.e., how far does X tend to be from its expectation?

- Mathematically,

$$\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

- 💡: Often, it's easier to use $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ and LOTUS to find $\mathbb{E}[X^2]$.

Definition 6 (Standard deviation). For a random variable X , $\text{SD}[X]$ is its *standard deviation*.

- Mathematically,

$$\text{SD}[X] = \sqrt{\text{Var}[X]}.$$

✎ Concept Checker 4. By definition, variance is the expected squared deviation of a random variable from its expectation—i.e., $\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$. Show that $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Solution

First, let $\mu = \mathbb{E}[X]$ to make the notation cleaner. Notice that μ is just a constant (since expectation returns a number as output), so linearity of expectation applies.

$$\begin{aligned} \text{Var}[X] &= \mathbb{E}[(X - \mu)^2] && \text{by definition} \\ &= \mathbb{E}[X^2 - 2X\mu + \mu^2] && \text{by FOIL} \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 && \text{by linearity} \\ &= \mathbb{E}[X^2] - 2\mu^2 + \mu^2 && \text{since } \mu = \mathbb{E}[X] \\ &= \mathbb{E}[X^2] - \mu^2 && \text{by simplifying} \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 && \text{since } \mu = \mathbb{E}[X] \end{aligned}$$

3.2 Variance Toolkit

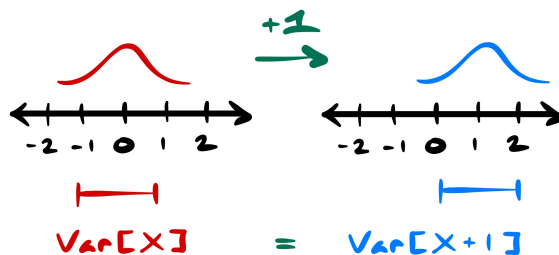


FIGURE 1: Shifting by a constant does not change $\text{Var}[X]$.

Definition 7 (Bilinearity of variance). For a random variable X and constants a and b ,

$$\text{Var}[aX + b] = a^2 \text{Var}[X].$$

Definition 8 (Sum of independent random variables). If $X_1, \dots, X_n$ are independent, then

$$\text{Var}[X_1 + \dots + X_n] = \text{Var}[X_1] + \dots + \text{Var}[X_n].$$

Definition 9 (Nonnegativity of variance). For a random variable X ,

$$\text{Var}[X] \geq 0.$$

4 Expectation and Variance of Famous Distributions

Idea. The *famous distributions* we've previously mentioned have known expectations and variances, which makes our lives easier!

Example 1 (Bernoulli). Let X be distributed *Bernoulli* with parameter p —i.e.,

$$X \sim \text{Bern}(p).$$

Then

$$\mathbb{E}[X] = p, \quad \text{Var}[X] = p(1 - p).$$

 **Concept Checker 5.** Let $X \sim \text{Bern}(p)$, with support $\{0, 1\}$. We showed that $\mathbb{E}[X] = p$.¹ Show that $\text{Var}[X] = p(1 - p)$.

Solution

$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. First, $\mathbb{E}[X] = p$, so $(\mathbb{E}[X])^2 = p^2$. Next, $\mathbb{E}[X^2] = \sum_{x \in \text{supp}(X)} x^2 P(X = x)$ by LOTUS. We can simplify this to $\mathbb{E}[X^2] = 0^2(1-p) + 1^2p = p$, so $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p)$. For some intuition, X can only crystallize to 0 or 1, and $0^2 = 0$ while $1^2 = 1$.

Example 2 (Binomial). Let X be distributed *Binomial* with parameters n and p —i.e.,

$$X \sim \text{Bin}(n, p).$$

Then

$$\mathbb{E}[X] = np, \quad \text{Var}[X] = np(1 - p).$$

 **Concept Checker 6.** Let $X \sim \text{Bin}(n, p)$. Show that $\mathbb{E}[X] = np$ and $\text{Var}[X] = np(1 - p)$.

¹See Definition 3.

Solution

By the story/property of the Binomial, $X = Y_1 + \dots + Y_n$, where $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \text{Bern}(p)$.

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}[Y_1 + \dots + Y_n] && \text{by substituting} \\ &= \mathbb{E}[Y_1] + \dots + \mathbb{E}[Y_n] && \text{by linearity} \\ &= p + \dots + p && \text{by expectation of Bernoulli} \\ &= np && \text{by simplifying}\end{aligned}$$

$$\begin{aligned}\text{Var}[X] &= \text{Var}[Y_1 + \dots + Y_n] && \text{by substituting} \\ &= \text{Var}[Y_1] + \dots + \text{Var}[Y_n] && \text{by sum of independent random variables} \\ &= p(1-p) + \dots + p(1-p) && \text{by variance of Bernoulli} \\ &= np(1-p) && \text{by simplifying}\end{aligned}$$

Example 3 (Geometric). Let X be distributed *Geometric* with parameter p —i.e.,

$$X \sim \text{Geom}(p).$$

Then

$$\mathbb{E}[X] = \frac{1-p}{p}, \quad \text{Var}[X] = \frac{1-p}{p^2}.$$

 **Concept Checker 7.** Let $X \sim \text{FS}(p)$. Find $\mathbb{E}[X]$ and $\text{Var}[X]$.

Solution

By the story/property of First Success, $X = Y + 1$, where $Y \sim \text{Geom}(p)$.

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}[Y + 1] && \text{by substituting} \\ &= \mathbb{E}[Y] + 1 && \text{by linearity} \\ &= \frac{1-p}{p} + 1 && \text{by expectation of Geometric} \\ &= \frac{1}{p} && \text{by simplifying}\end{aligned}$$

$$\begin{aligned}\text{Var}[X] &= \text{Var}[Y + 1] && \text{by substituting} \\ &= \text{Var}[Y] && \text{by bilinearity} \\ &= \frac{1-p}{p^2} && \text{by variance of Geometric}\end{aligned}$$

Example 4 (Negative Binomial). Let X be distributed *Negative Binomial* with parameters r and p —i.e.,

$$X \sim \text{NBin}(r, p).$$

Then

$$\mathbb{E}[X] = r \left(\frac{1-p}{p} \right), \quad \text{Var}[X] = r \left(\frac{1-p}{p^2} \right).$$

Example 5 (Hypergeometric). Let X be distributed *Hypergeometric* with parameters w , b , and n —i.e.,

$$X \sim \text{HGeom}(w, b, n).$$

Then

$$\mathbb{E}[X] = \frac{nw}{b+w}, \quad \text{Var}[X] = \left(\frac{nbw}{(w+b)^2} \right) \left(\frac{w+b-n}{w+b-1} \right).$$

5 Poisson Distribution

Idea. The final discrete distribution we'll introduce is the *Poisson* distribution, which is unique in that its story is a bit complicated. However, many real-world phenomena are modeled using the Poisson as an *approximation*.

5.1 Fundamentals

Definition 10 (Poisson). Let X be distributed *Poisson* with parameter λ —i.e.,

$$X \sim \text{Pois}(\lambda).$$

- **Story:** X is the number of occurrences for a rare event with rate λ in t units of time/space (with $t = 1$ as default).

- **Example:** Let X be the number of calls received in an hour at a customer service desk, which receives an average of 4 calls per hour. Then $X \sim \text{Pois}(4)$.

- **PMF and Support:**

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

for $x \in \{0, 1, \dots\}$.

- **Properties:** If $X \sim \text{Pois}(\lambda_1)$ and $Y \sim \text{Pois}(\lambda_2)$ with $X \perp\!\!\!\perp Y$, then

$$X + Y \sim \text{Pois}(\lambda_1 + \lambda_2)$$

and

$$X \mid \{X + Y = n\} \sim \text{Bin}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right).$$

 **Concept Checker 8.** If X is the number of calls received in an hour and $X \sim \text{Pois}(4)$, then what is the distribution of the number of calls received in *two* hours?

Solution

Let Y be the number of calls received in two hours. $Y \sim \text{Pois}(8)$ since the rate is 4 calls in 1 unit of time, and we're now concerned with $t = 2$.

Definition 11 (Connection between Poisson and Binomial). If $X \sim \text{Bin}(n, p)$ and we let $n \rightarrow \infty$ and $p \rightarrow 0$ such that $\lambda = np$ remains fixed, then the PMF of X converges to the PMF of $\text{Pois}(\lambda)$.

Definition 12 (Poisson paradigm). Let $A_1, \dots, A_n$ be independent (or weakly dependent) events, where $p_i = P(A_i)$. If $X = \sum_{i=1}^n \mathbf{1}\{A_i\}$ is the number of events that occur (with n large and p_i small), then X is *approximately* distributed $\text{Pois}(\lambda)$, where $\lambda = \sum_{i=1}^n p_i$.

- We denote this *approximate distribution* as $X \dot{\sim} \text{Pois}(\lambda)$.

Definition 13 (Chicken-egg story). Suppose that a chicken lays N eggs, where $N \sim \text{Pois}(\lambda)$. Each egg independently hatches with probability p and fails to hatch with probability $q = 1 - p$. Let X be the number of eggs that hatch and Y be the number of eggs that don't hatch (such that $X + Y = N$). The *joint PMF* of X and Y , $P(X = x, Y = y)$, is the product of two terms, where one relies only on x and the other relies only on y —i.e.,

$$P(X = x, Y = y) = \left(\frac{e^{-\lambda p} (\lambda p)^x}{x!} \right) \left(\frac{e^{-\lambda q} (\lambda q)^y}{y!} \right) = P(X = x)P(Y = y).$$

- Mathematically, if $N \sim \text{Pois}(\lambda)$ and $X \mid \{N = n\} \sim \text{Bin}(n, p)$, then

$$X \sim \text{Pois}(\lambda p)$$

and

$$Y = N - X \sim \text{Pois}(\lambda q),$$

with

$$X \perp\!\!\!\perp Y.$$

-  Often, we apply the chicken-egg story when $N \sim \text{Pois}(\lambda)$ is split into 2 groups: successes (with probability p) and failures (with probability $q = 1 - p$).

 **Concept Checker 9.** In a certain election, there are $V \sim \text{Pois}(\lambda)$ registered voters. Each registered voter is a Democrat with probability p and a Republican with probability $1 - p$, independent of other voters. Each registered voter actually votes in the election with probability s and stays home with probability $1 - s$, independent of other voters and independent of party affiliations. Let X be the number of registered Democrats who actually vote in the election. What is the distribution of X ?²

Solution

$X \sim \text{Pois}(\lambda ps)$ by chicken-egg story. Notice that we're “splitting” the Poisson random variable V .

5.2 Expectation and Variance of Poisson

Example 6 (Poisson). Let X be distributed *Poisson* with parameter λ —i.e.,

$$X \sim \text{Pois}(\lambda).$$

Then

$$\mathbb{E}[X] = \lambda, \text{ Var}[X] = \lambda.$$

²Inspired by Problem 2 in “Practice Midterm 2” by Joseph K. Blitzstein.

6 Table of Distributions

Idea. We conclude with a review of the famous distributions we've covered so far.

Name	Story	PMF & Support	Expectation	Variance	Properties
$X \sim \text{Bern}(p)$	X succeeds (i.e., $X = 1$) with probability p and fails (i.e., $X = 0$) with probability $1 - p$	$P(X = x) = p^x(1 - p)^{1-x}$ for $x \in \{0, 1\}$	p	$p(1 - p)$	
$X \sim \text{Bin}(n, p)$	X is the number of successes in n independent trials that each have success probability p	$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$ for $x \in \{0, 1, \dots, n\}$	np	$np(1 - p)$	If $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ with $X \perp\!\!\!\perp Y$, then $X + Y \sim \text{Bin}(n + m, p)$; If $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bern}(p)$, then $X_1 + \dots + X_n \sim \text{Bin}(n, p)$
$X \sim \text{Geom}(p)$	X is the number of failures until we observe a success in independent trials that each have success probability p , <i>not</i> counting that first success	$P(X = x) = (1 - p)^x p$ for $x \in \{0, 1, \dots\}$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$	If $X \sim \text{Geom}(p)$, then $X + 1 \sim \text{FS}(p)$
$X \sim \text{NBin}(r, p)$	X is the number of failures until we observe the r th success in independent trials that each have success probability p , <i>not</i> counting any successes	$P(X = x) = \binom{x+r-1}{r-1} p^r (1 - p)^x$ for $x \in \{0, 1, \dots\}$	$r \left(\frac{1-p}{p} \right)$	$r \left(\frac{1-p}{p^2} \right)$	If $X_1, \dots, X_r \stackrel{\text{i.i.d.}}{\sim} \text{Geom}(p)$, then $X_1 + \dots + X_r \sim \text{NBin}(r, p)$
$X \sim \text{HGeom}(w, b, n)$	X is the number of <i>desirable</i> objects in a sample if we draw n objects, <i>without replacement</i> , from a population with w <i>desirable</i> objects and b <i>undesirable</i> objects (with $w + b$ total objects)	$P(X = x) = \frac{\binom{w}{x} \binom{b}{n-x}}{\binom{w+b}{n}}$ for $x \in \{\max(0, n - b), \dots, \min(n, w)\}$	$\frac{nw}{b+w}$	$\left(\frac{nw}{(w+b)^2} \right) \left(\frac{w+b-n}{w+b-1} \right)$	If $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ with $X \perp\!\!\!\perp Y$, then $X \{X + Y = r\} \sim \text{HGeom}(n, m, r)$
$X \sim \text{Pois}(\lambda)$	X is the number of occurrences for a rare event with rate λ in t units of time/space (with $t = 1$ as default)	$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$ for $x \in \{0, 1, \dots\}$	λ	λ	If $X \sim \text{Pois}(\lambda_1)$ and $Y \sim \text{Pois}(\lambda_2)$ with $X \perp\!\!\!\perp Y$, then $X + Y \sim \text{Pois}(\lambda_1 + \lambda_2)$ and $X \{X + Y = n\} \sim \text{Bin}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right)$

7 Exercises

 **Problem 1.** Suppose that Alice is taking a multiple choice exam that consists of 20 questions. Each question has 5 options but only 1 correct answer. Unfortunately, Alice forgets to study and randomly guesses for each question.

- What is the probability that Alice answers all questions correctly?
- What is the expected number of questions Alice answers correctly?
- For this part only, suppose that the exam consists of *checkbox* questions. Each question has 5 options, and any number of them (from 0 to 5) may be correct. Alice answers each question by randomly choosing which options to check, and she answers the question correctly only if she checks all the correct options (and nothing more). A passing grade requires at least 75% of the answers to be correct. What is the probability that Alice gets a passing grade? You may leave your answer as a sum.
- For this part only, suppose that the exam progresses in difficulty, with i options for question $i \in \{1, \dots, 20\}$ (e.g., question 4 has 4 options, question 5 has 5 options, and so

on). Like before, each question has only 1 correct answer. What is the expected number of questions Alice answers correctly? You may leave your answer as a sum.

Solution

(a) Let Q_i be the event that Alice answers question i correctly. By naive probability, $P(Q_i) = \frac{1}{5}$. Importantly, since Alice randomly guesses for each question, $Q_i \perp \perp Q_j \forall i \neq j$. By the multiplication rule, $P(\text{Alice answers all questions correctly}) = \left(\frac{1}{5}\right)^{20}$, which is very small!

(b) Let X be the number of questions Alice answers correctly. Notice that $X = \mathbb{1}\{Q_1\} + \dots + \mathbb{1}\{Q_{20}\}$, and $\mathbb{1}\{Q_i\} \stackrel{\text{i.i.d.}}{\sim} \text{Bern}\left(\frac{1}{5}\right)$. By the story of the Binomial, $X \sim \text{Bin}\left(20, \frac{1}{5}\right)$, so $\mathbb{E}[X] = (20)\left(\frac{1}{5}\right) = 4$.

(c) By the multiplication rule, for each question, there are $2^5 = 32$ possible subsets/ways to check the options (we can either check or not check each of the 5 options), but only 1 of those ways corresponds to the correct answer. By naive probability, $P(Q_i) = \frac{1}{32}$, so $X \sim \text{Bin}\left(20, \frac{1}{32}\right)$.

$$\begin{aligned}
 P(\text{Alice gets a passing grade}) &= P(X \geq 15) && \text{by definition of } X \\
 &= \sum_{x=15}^{20} P(X = x) && \text{since } X \text{ is discrete} \\
 &= \sum_{x=15}^{20} \binom{20}{x} \left(\frac{1}{32}\right)^x \left(\frac{31}{32}\right)^{20-x} && \text{by Binomial PMF}
 \end{aligned}$$

(d) Importantly, X is no longer Binomial since each question has a different success probability. By naive probability, $P(Q_i) = \frac{1}{i}$.

$$\begin{aligned}
 \mathbb{E}[X] &= \mathbb{E}\left[\sum_{i=1}^{20} \mathbb{1}\{Q_i\}\right] && \text{since } X = \sum_{i=1}^{20} \mathbb{1}\{Q_i\} \\
 &= \sum_{i=1}^{20} \mathbb{E}[\mathbb{1}\{Q_i\}] && \text{by linearity} \\
 &= \sum_{i=1}^{20} P(Q_i) && \text{by fundamental bridge} \\
 &= \sum_{i=1}^{20} \frac{1}{i} && \text{by naive probability} \\
 &= 3.59773 && \text{by simplifying}
 \end{aligned}$$